16

Introduction to Glomerular Disease: Clinical Presentations

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DEFINITION

Glomerular disease has clinical presentations that vary from the asymptomatic individual who is found to have hypertension, edema, hematuria, or proteinuria at a routine medical assessment to a patient who has fulminant illness with acute kidney injury (AKI) possibly associated with life-threatening extrarenal disease (Fig. 16.1). The most dramatic symptomatic presentations are uncommon. Asymptomatic urine abnormalities are much more common but may also indicate a wide range of nonglomerular urinary tract diseases.

CLINICAL EVALUATION OF GLOMERULAR DISEASE

The history, physical examination, and diagnostic investigations are aimed at excluding nonglomerular disease, finding evidence of associated multisystem disease, and establishing kidney function.

History

The majority of glomerular diseases do not lead to symptoms that patients will report. However, specific questioning may reveal edema, hypertension, foamy urine, or urinary abnormalities noted during routine testing (e.g., during scheduled medical examinations). Multisystem diseases associated with glomerular disease include diabetes, hypertension, amyloid, lupus, and vasculitis. Classic genetic causes of kidney disease may include Alport syndrome, especially if associated with hearing loss (see Chapter 48); familial forms of immunoglobulin A (IgA) nephropathy (see Chapter 24); focal segmental glomerulosclerosis (FSGS) secondary to mutations in podocin or other molecules involved in glomerular permeability (see Chapters 19 and 20); complement-mediated glomerulonephritis (GN) (see Chapter 23); thrombotic microangiopathies (see Chapter 30); and other rare conditions (see Chapter 29). Morbid obesity can be associated with FSGS. Certain drugs and toxins may cause glomerular disease, including nonsteroidal antiinflammatory drugs (NSAIDs) and interferon in minimal change disease (MCD); penicillamine, NSAIDs, and mercury (e.g., skin-lightening creams) in membranous nephropathy; pamidronate and heroin in FSGS; smoking in nodular glomerulosclerosis; and cyclosporine, tacrolimus, mitomycin C, and oral contraceptives in hemolytic uremic syndrome (HUS). Recent or persistent infection, especially streptococcal or staphylococcal infection, endocarditis, and certain viral infections (see Chapters 22, 57, and 58) also may be associated with a variety of glomerular diseases.

Malignancies associated with glomerular disease include lung, breast, and gastrointestinal (GI) adenocarcinoma in membranous nephropathy; Hodgkin disease in MCD; non-Hodgkin lymphoma in membranoproliferative GN; and renal cell carcinoma in amyloid disease (see Chapter 28). Kidney disease is occasionally the first manifestation of a tumor.

Physical Examination

The presence of dependent pitting edema suggests nephrotic syndrome, heart failure, or cirrhosis. In the nephrotic patient, edema is often periorbital in the morning (Fig. 16.2), whereas the face is not affected overnight in edema associated with heart failure (edema distributes by gravity, and patients with heart failure often cannot lie flat due to orthopnea or cirrhosis (because of pressure on the diaphragm from ascites). Severe nephrosis can lead to edema of genitals and the abdominal wall, ascites, and pleural effusions. Edema is unpleasant, leading to feelings of tightness in the limbs and a bloated abdomen, with practical problems of clothes and shoes no longer fitting. Surprisingly, however, edema may become massive in nephrotic syndrome before patients seek medical help; fluid gains of 20 kg (44 lb) or more are not unusual (Fig. 16.3). The edema becomes firm and stops pitting only when it is long-standing. In children, fluid retention may be striking with acute GN (nephritic syndrome). Chronic hypoalbuminemia is also associated with white nails or white bands if nephrotic syndrome is transient (Muehrcke lines; Fig. 16.4). Xanthelasmas may be present as a result of the hyperlipidemia associated with long-standing nephrotic syndrome (Fig. 16.5).

Pulmonary signs suggest one of the so-called pulmonary-renal syndromes (see Boxes 25.3 and 25.4). Palpable purpura may be seen in vasculitis, systemic lupus, cryoglobulinemia, or endocarditis.

Laboratory Studies

Assessment of kidney function and careful examination of the urine are critical (see Chapters 3 and 4). The quantity of urine protein and the presence or absence of dysmorphic red cells and casts help classify the clinical presentation (see Fig. 16.1).

Useful serologic tests include antiphospholipase A2 receptor and other autoantibodies (see Chapter 21) for membranous GN, antinuclear and anti-DNA antibodies for lupus, cryoglobulins and rheumatoid factor suggesting cryoglobulinemia, anti–glomerular basement membrane (anti-GBM) antibodies for Goodpasture disease, antineutrophil cytoplasmic autoantibody (ANCA) for vasculitis, and

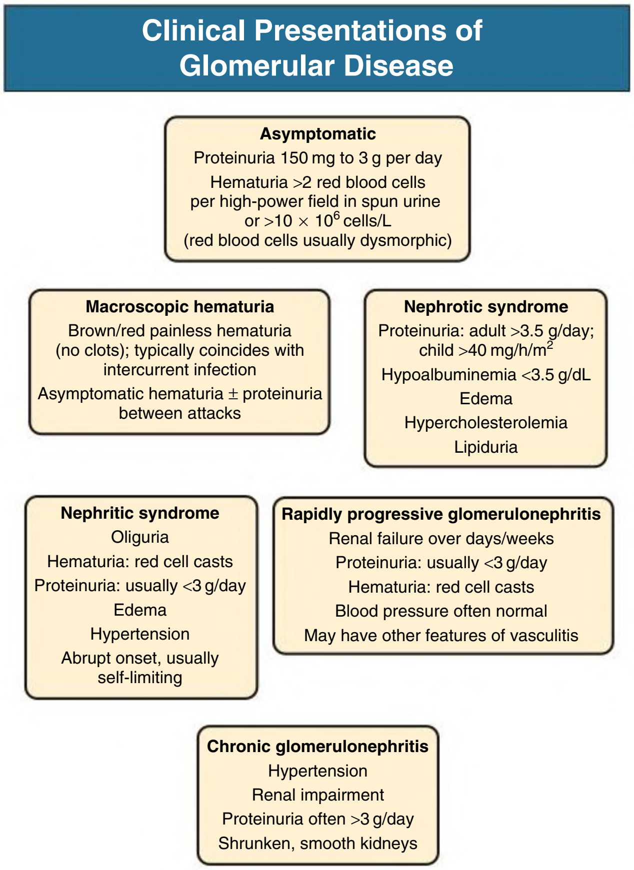

Fig. 16.1 Clinical presentation of various glomerular diseases.

Fig. 16.2 Nephrotic Edema. Periorbital edema in the early morning in a nephrotic child. The edema resolves during the day under the influence of gravity.

Fig. 16.3 Nephrotic Edema. Severe peripheral edema in nephrotic syndrome; note the blisters caused by intradermal fluid.

Fig. 16.4 Muehrcke Lines (Bands) in Nephrotic Syndrome. The white line grew during a transient period of hypoalbuminemia caused by nephrotic syndrome.

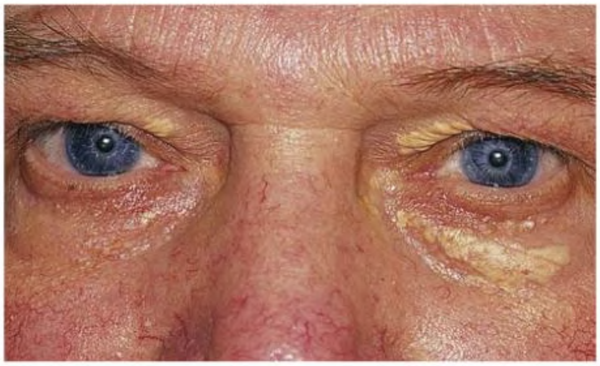

Fig. 16.5 Xanthelasmas in Nephrotic Syndrome. These prominent xanthelasmas developed within 2 months in a patient with recent onset of severe nephrotic syndrome and serum cholesterol level of 550 mg/dL (14.2 mmol/L).

antistreptolysin O titer or streptozyme test for poststreptococcal GN. Serum and urine electrophoresis will detect monoclonal light chains or heavy chains, and assays for free light chains in serum or urine may aid in their quantification, as in myeloma-associated amyloid or light-chain deposition disease.

Blood cultures and testing for hepatitis B, hepatitis C, and human immunodeficiency virus (HIV) infection are also useful.

Measurement of systemic complement pathway activation by testing for serum C3, C4, and CH50 (50% hemolyzing dose of complement) is often helpful in limiting the differential diagnosis (Table 16.1).

The importance of genetic evaluation in patients with GN is discussed in Chapters 20 and 23.

TABLE 16.1 Hypocomplementemia in Glomerular Disease

Pathways Affected	Complement Changes	Glomerular Disease	Nonglomerular Disease
Classical pathway activation	C3↓, C4↓, CH50↓	Lupus nephritis (especially class IV) Cryoglobulinemia Membranoproliferative GN type 1	
Alternative pathway activation	C3↓, C4 normal, CH50↓	Poststreptococcal GN GN associated with other infection[a] (e.g., endocarditis, shunt nephritis) HUS	Atheroembolic renal disease
	plus C3 nephritic factor	Dense deposit disease	
Reduced complement synthesis	Acquired		Hepatic disease Malnutrition
	Hereditary C2 or C4 deficiency Factor H deficiency	Lupus nephritis Familial HUS Dense deposit disease	

[a]GN with visceral abscesses is generally associated with normal or increased complement (elevations occur because complement components are acute-phase reactants).
CH50, 50% hemolyzing dose of complement; *GN,* glomerulonephritis; *HUS,* hemolytic uremic syndrome.

Imaging

A kidney ultrasound is recommended in the workup to ensure the presence of two kidneys, to rule out obstruction or anatomic abnormalities, and to assess kidney size. Kidney size is often normal in GN, although large kidneys (>14 cm) are sometimes seen in nephrotic syndrome associated with diabetes, amyloid disease, or HIV infection. Large kidneys also can occasionally be seen with any acute severe GN and acute interstitial nephritis. Small kidneys (<9 cm) and/or severe cortical thinning suggests advanced chronic kidney disease (CKD) and should limit enthusiasm for kidney biopsy or aggressive immunosuppressive therapies.

Kidney Biopsy

Kidney biopsy is generally required to establish the type of glomerular disease and to guide treatment decisions (see Chapter 7). In some patients, however, kidney biopsy is not performed. If nephrotic children (ages 2–12) have no unusual clinical features, the probability of MCD is so high that corticosteroids can be initiated without biopsy (see Chapter 18). In patients with acute nephritic syndrome, if all features point to poststreptococcal GN, especially in an epidemic, biopsy can be reserved for those without early spontaneous improvement (see Chapter 57). In Goodpasture disease (see Chapter 25), the presence of lung hemorrhage and rapidly progressive kidney failure with urinary red cell casts and high levels of circulating anti-GBM antibody establishes the diagnosis without the need for a biopsy. In patients with systemic features of vasculitis, a positive ANCA titer, negative blood cultures, and a biopsy specimen from another site showing vasculitis are sufficient to secure a diagnosis of renal vasculitis. However, even when kidney biopsy is not needed for diagnosis, it may provide important clues to disease activity and chronicity. Biopsy is also not generally performed in patients with long-standing diabetes with characteristic findings suggestive of diabetic nephropathy and other evidence of microvascular complications of diabetes (see Chapter 31). Biopsy may not be indicated in many patients with glomerular disease presenting with minor, asymptomatic urine abnormalities and well-preserved kidney function because the prognosis is excellent and histologic findings will not alter management.

ASYMPTOMATIC URINE ABNORMALITIES

The random nature of urine testing in most communities inevitably means that much mild glomerular disease remains undetected. In some countries, symptomless individuals may have a urine test only if they require medical approval for some key life event, such as obtaining life insurance, joining the armed forces, or sometimes for employment purposes. In other countries, such as Japan, urinalysis is performed routinely in school or for employment. These different practices may partly account for the apparently variable incidence of certain diseases, such as IgA nephropathy, which often manifests as asymptomatic proteinuria and microhematuria. Asymptomatic low-grade proteinuria and microhematuria and the combination of the two also increase in prevalence with age[1] (Fig. 16.6). Nevertheless, there is no evidence to justify routine population-wide screening for asymptomatic urine abnormalities as kidney biopsy and therapeutic intervention are rarely required when kidney function is preserved. For certain high-risk populations, such as patients with diabetes or hypertension, searching for albuminuria ("case-finding") may be useful as it carries increased risk for cardiovascular disease.

Asymptomatic Microhematuria

Microhematuria is defined as the presence of more than two red blood cells (RBCs) per high-power field in a spun urine sediment (3000 rpm for 5 minutes) or more than 10×10^6 RBCs/L. Microhematuria is common in many glomerular diseases, especially IgA nephropathy and thin basement membrane nephropathy, although there are many other causes of hematuria (see Chapters 48 and 63). A glomerular origin should be assumed if more than 5% of RBCs are acanthocytes or dysmorphic (see Chapter 4) or if the hematuria is accompanied by RBC casts or proteinuria (Fig. 16.7).

Pathogenesis

Glomerular hematuria results from small breaks in the GBM that allow extravasation of RBCs into the urinary space. This may occur in the peripheral capillary wall but more often occurs in the paramesangial basement membrane, particularly in diseases involving injury to the mesangium (mesangiolysis). As long as the kidney tubules are intact, low amounts of serum proteins lost together with RBCs in damaged glomeruli can be fully reabsorbed, resulting in "isolated" microhematuria.

Evaluation

The evaluation of microhematuria, discussed further in Chapters 48 and 63, begins with a thorough history. Urine culture should

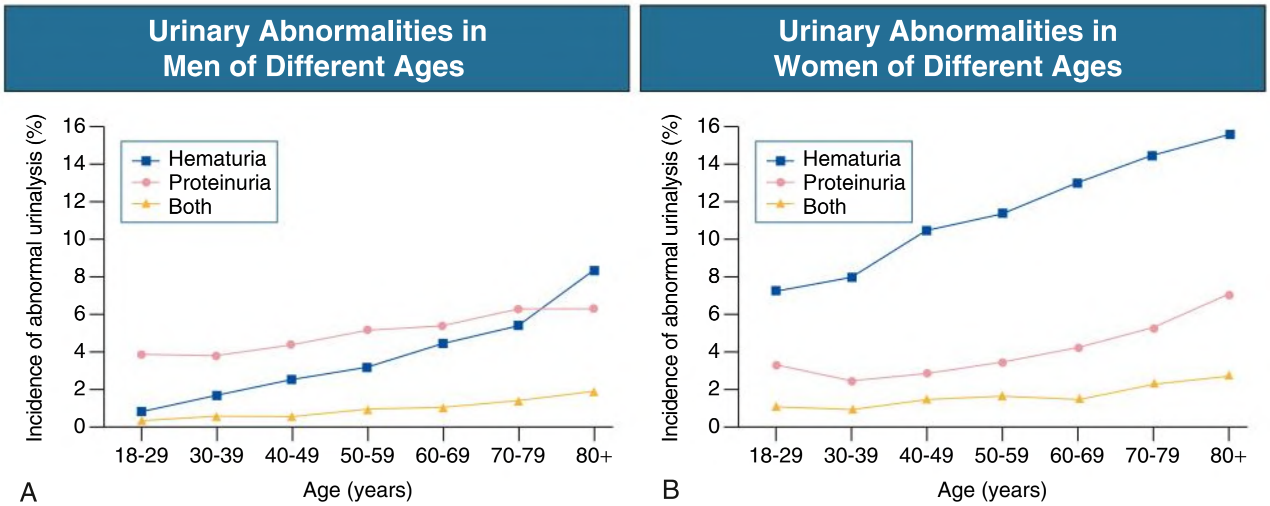

Fig. 16.6 **Prevalence of Asymptomatic Proteinuria and Hematuria With Age.** Mass screening of a population of 107,192 adult men (A) and women (B) in Okinawa, Japan. Hematuria is more common in women. (Modified from Iseki K, Iseki C, Ikemiya Y, Fukiyama K. Risk of developing end-stage renal disease in a cohort of mass screening. *Kidney Int.* 1996;49:800–805.)

Fig. 16.7 A red blood cell cast typical of glomerular hematuria.

exclude urinary or prostatic infection. Phase contrast microscopy should follow in cases of persistent microhematuria to search for dysmorphic RBCs and RBC casts. In the absence of urine infection or a bladder catheter, any detectable albuminuria (>0.3 g/24 h) in the patient with microhematuria virtually excludes "urologic" bleeding and strongly suggests a glomerular origin. If this evaluation is nondiagnostic, kidney imaging is performed to exclude anatomic lesions such as stones, tumors, polycystic kidneys, or arteriovenous malformations.

In individuals older than 40 years with persistent isolated microhematuria without evidence of a glomerular origin (see previous discussion), cystoscopy is mandatory to exclude uroepithelial malignant disease. In people younger than 40 years, malignancy is so rare that cystoscopy is not recommended. If all the prior study results are normal, a glomerular cause is likely.[2] The glomerular cause can be determined only by kidney biopsy, but this is rarely done because the prognosis is excellent in patients with normal kidney function, normal blood pressure, and low-grade proteinuria (<0.5 g/day). However, repeated evaluation and prolonged follow-up are mandatory for as long as the urinary abnormality persists.

Asymptomatic Nonnephrotic Proteinuria

The hallmark of glomerular disease is the excretion of protein in the urine. Normal urine protein excretion is less than 150 mg/24 h, consisting of 20 to 30 mg of albumin, 10 to 20 mg of low-molecular-weight proteins that undergo glomerular filtration, and 40 to 60 mg of secreted proteins (e.g., Tamm-Horsfall, IgA). Proteinuria is identified and quantified by dipstick testing or by assay in timed urine collections (see Chapter 4).

An albumin excretion rate of 30 to 300 mg of albumin per day (previously termed *microalbuminuria*), equivalent to a urine albumin-to-creatinine (gram/gram) ratio of 0.03 to 0.3, is detected by quantitative immunoassay or by special urine dipsticks because this is below the sensitivity of the normal dipstick (see Chapter 31). This measurement is primarily used to identify diabetic individuals at risk for development of nephropathy and to assess cardiovascular risk, for example, in patients with hypertension.

Nonnephrotic proteinuria is usually defined as a urine protein excretion of less than 3.5 g/24 h or a urine protein/creatinine ratio of less than 3 g/g. Whereas nephrotic-range proteinuria is absolutely characteristic of glomerular disease, lower levels of proteinuria (<3.5 g/24 h) are much less specific and may occur with a wide range of nonglomerular parenchymal diseases as well as other conditions that must be excluded by clinical evaluation and investigation.

Increased urine protein excretion may result from alterations in glomerular permeability or tubulointerstitial disease, although only in glomerular disease will it be in the nephrotic range. Protein excretion also can increase if there is greater filtration through normal glomeruli (overflow proteinuria).

Overflow Proteinuria

Overflow proteinuria is typical of urinary light chain excretion. It is seen in myeloma but can occur in other settings (e.g., release of lysozyme by leukemic cells) and should be suspected when the urine dipstick is negative for albumin despite detection of large amounts of proteinuria by other tests.

Tubular Proteinuria

Tubulointerstitial disease can be associated with low-grade proteinuria (usually <2 g/day). In addition to the loss of tubular proteins such as α_1- or β_2-microglobulin, there will also be some albuminuria secondary to impaired tubular reabsorption of filtered albumin. Tubular proteinuria accompanying glomerular proteinuria is an adverse prognostic sign in various glomerular diseases because it usually indicates advanced tubulointerstitial damage.

Glomerular Proteinuria

Glomerular proteinuria is further classified into transient or hemodynamic (functional) proteinuria, proteinuria that is present only during the day (orthostatic), and persistent or fixed proteinuria.

Functional proteinuria. Functional proteinuria refers to the transient nonnephrotic proteinuria that can occur with fever, exercise, heart failure, and hyperadrenergic or hyperreninemic states. Functional proteinuria is benign, usually assumed to be hemodynamic in origin, and the result of increases in single-nephron flow or pressure.

Orthostatic proteinuria. In children and young adults, low-grade glomerular proteinuria may be orthostatic, meaning that proteinuria is absent when urine is generated in the recumbent position. If there is no proteinuria in early-morning urine, the diagnosis of orthostatic proteinuria can be made. In patients with fixed orthostatic proteinuria, renal plasma flow and glomerular filtration rate (GFR) decrease in the upright position because of a lower systemic blood pressure. As recently proposed, the decreased GFR translates into a lower *streaming potential* across the filtration barrier, which under physiologic conditions retains albumin within the blood. When GFR and filtration pressure are reduced beyond a certain threshold, albumin is no longer excluded efficiently from the filter by electrophoresis and consequently leaks through the filter, explaining reversible low-grade proteinuria in the upright position in these patients.[3]

Total urine protein in the patient with orthostatic proteinuria is usually less than 1 g/24 h; hematuria and hypertension are absent. Kidney biopsy usually shows normal morphology or occasionally mild glomerular change. The prognosis is uniformly good, and kidney biopsy is not indicated.[4]

Fixed nonnephrotic proteinuria. Fixed nonnephrotic proteinuria is usually caused by glomerular disease. If GFR is preserved and proteinuria is less than 0.5 to 1 g/day, biopsy is not indicated but prolonged follow-up is necessary if significant proteinuria persists, to rule out the possibility of disease progression. Previous studies indicate that the biopsy findings in these patients can be similar to those seen in nephrotic syndrome (most commonly FSGS or membranous nephropathy), although milder lesions are more common, particularly mesangial proliferative GN or IgA nephropathy. In general, other than regular monitoring and blood pressure control as needed, no treatment is necessary.

While controversial, some nephrologists perform a kidney biopsy in patients with normal GFR if nonnephrotic proteinuria exceeds 1 g/day, in particular if it persists after initiation of angiotensin-converting enzyme (ACE) inhibitor or angiotensin receptor blocker (ARB) therapy.

Fig. 16.8 summarizes the evaluation of isolated asymptomatic proteinuria.

Asymptomatic Proteinuria With Hematuria

Patients with coincident asymptomatic hematuria and proteinuria have a much greater risk for significant glomerular injury, hypertension, and progressive kidney dysfunction. Minor histologic changes are less common. Kidney biopsy is often performed even if urine protein is only 0.5 to 1 g/24 h if there is also persistent microhematuria with casts and/or declining kidney function.

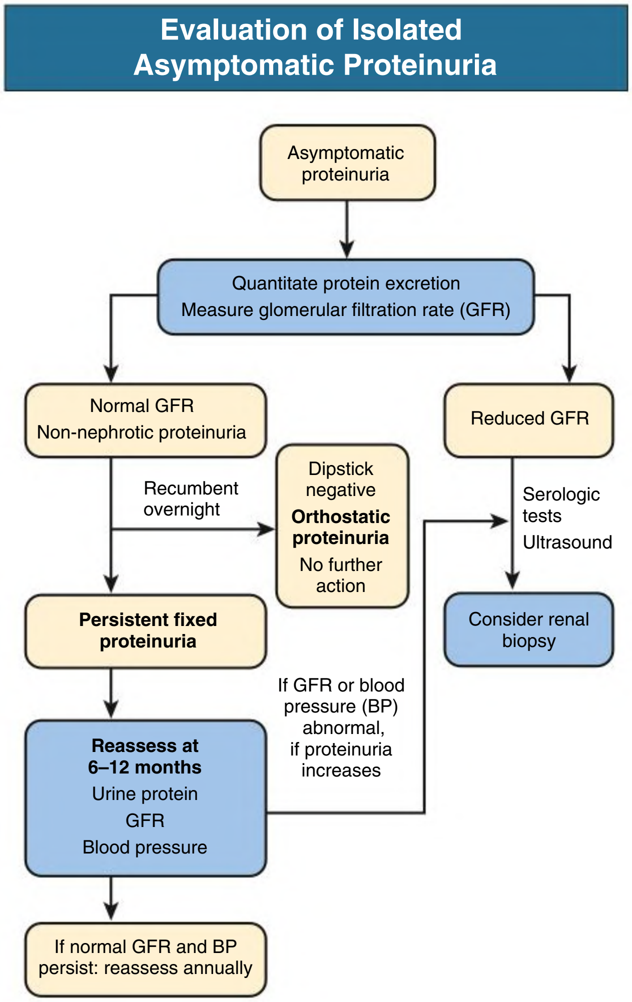

Fig. 16.8 Evaluation of patients with isolated asymptomatic proteinuria.

MACROHEMATURIA

Episodic painless macrohematuria associated with glomerular disease is often brown or “smoky” rather than red, and clots are unusual. Macrohematuria must be distinguished from other causes of red or brown urine, including hemoglobinuria, myoglobinuria, porphyrias, consumption of food dyes (particularly beetroot), and intake of drugs (especially rifampin/rifampicin).

Macrohematuria caused by glomerular disease is seen mainly in children and young adults and is rare after age 40 years. Most cases are caused by IgA nephropathy, but hematuria may occur with other glomerular and nonglomerular kidney diseases, including acute interstitial nephritis. Although macrohematuria is typically painless, the patient may have an accompanying dull loin ache that suggests other diagnoses, such as nephrolithiasis or loin-pain hematuria syndrome (see Chapter 60). In IgA nephropathy, the frank hematuria is usually episodic, occurring within a day of an upper respiratory tract infection. There is a clear distinction between this history and the 2- to 3-week latency between an upper respiratory tract infection and hematuria that is highly suggestive of postinfectious (usually poststreptococcal) GN; furthermore, patients with poststreptococcal disease usually will have other features of nephritic syndrome.

Macrohematuria requires urologic evaluation, including cystoscopy, at any age unless the history is characteristic of glomerular hematuria.

TABLE 16.2 Common Glomerular Diseases Presenting as Nephrotic Syndrome in Adults

Disease	Associations	Serologic Tests
MCD	Allergy, atopy, NSAIDs, Hodgkin disease	None*
FSGS	African American race	—
	HIV infection	HIV antibody
	Heroin, pamidronate	—
MN	Idiopathic drugs: gold, penicillamine, NSAIDs	Antibodies to PLA2R and other new autoantigens (see Chapter 21)
	Infections: hepatitis B and C; malaria	Hepatitis B surface antigen, anti–hepatitis C virus antibody
	Lupus nephritis	Anti-DNA antibody
	Malignancy: breast, lung, gastrointestinal tract	—
MPGN type I	C4 nephritic factor	C3↓, C4↓
Dense deposit disease	C3 nephritic factor	C3↓, C4 normal
Cryoglobulinemic MPGN	Hepatitis C	Anti–hepatitis C virus antibody, rheumatoid factor, C3↓, C4↓, CH50↓
Amyloid disease	Myeloma	Plasma free light chains
	Rheumatoid arthritis, bronchiectasis, Crohn disease (and other chronic inflammatory conditions), familial Mediterranean fever	Serum protein electrophoresis, urine immunoelectrophoresis C-reactive protein
Diabetic nephropathy	Other diabetic microangiopathy	None

*Possibly anti-nephrin antibodies in the future.
CH50, 50% Hemolyzing dose of complement; *FSGS,* focal segmental glomerulosclerosis; *HIV,* human immunodeficiency virus; *MCD,* minimal change disease; *MN,* membranous nephropathy; *MPGN,* membranoproliferative glomerulonephritis; *NSAIDs,* nonsteroidal antiinflammatory drugs; *PLA_2R,* phospholipase A_2 receptor.

TABLE 16.3 Age-Related Variations in the Prevalence (%) of Nephrotic Syndrome

		YOUNG ADULT		MIDDLE AND OLD AGE	
	Child (<15 yr)	Whites	Blacks	Whites	Blacks
Minimal change disease	78	23	15	21	16
Focal segmental glomerulosclerosis	8	19	55	13	35
Membranous nephropathy	2	24	26	37	24
Membranoproliferative glomerulonephritis	6	13	0	4	2
Other glomerulonephritides	6	14	2	12	12
Amyloid	0	5	2	13	11

Data from Cameron JS. Nephrotic syndrome in the elderly. *Semin Nephrol.* 1996;16:319–329; and Haas M, Meehan SM, Karrison TG, Spargo BH. Changing etiologies of unexplained adult nephrotic syndrome: A comparison of renal biopsy findings from 1976–1979 and 1995–1997. *Am J Kidney Dis.* 1997;30:621–631.

NEPHROTIC SYNDROME

Definition

Nephrotic syndrome is pathognomonic of glomerular disease. It is a clinical syndrome with a characteristic pentad[5] (see Fig. 16.1). Patients may be nephrotic with preserved renal function, but in many, progressive kidney failure will become superimposed when nephrotic syndrome is prolonged.

Independent of the risk for progressive loss of kidney function, the nephrotic syndrome has far-reaching effects on the general health of the patient. Some episodes of nephrotic syndrome are self-limited, and a few patients respond completely to specific treatment (e.g., corticosteroids in MCD), but for most patients it is a relapsing or chronic condition. Not all patients with proteinuria above 3.5 g/24 h will have full nephrotic syndrome; some have a normal serum albumin concentration and no edema. This difference presumably reflects the varied response of protein metabolism; some patients sustain an increase in albumin synthesis in response to heavy proteinuria that may even normalize serum albumin.

Etiology

Table 16.2 shows the major causes of nephrotic syndrome. Proteinuria in the nephrotic range in the absence of edema and hypoalbuminemia has similar causes. The relative frequency of the different glomerular diseases varies with age (Table 16.3). Although it is predominant in childhood, MCD remains common at all ages.[6] The prevalence of FSGS in African Americans is increased, which may explain why FSGS is becoming more common in US adults but not in European adults.[7,8]

Hypoalbuminemia

Hypoalbuminemia is mainly a consequence of urinary losses. The liver responds by increasing albumin synthesis, but this compensatory mechanism appears to be blunted in nephrotic syndrome.[9] The end result is that serum albumin falls further. White bands in the nails

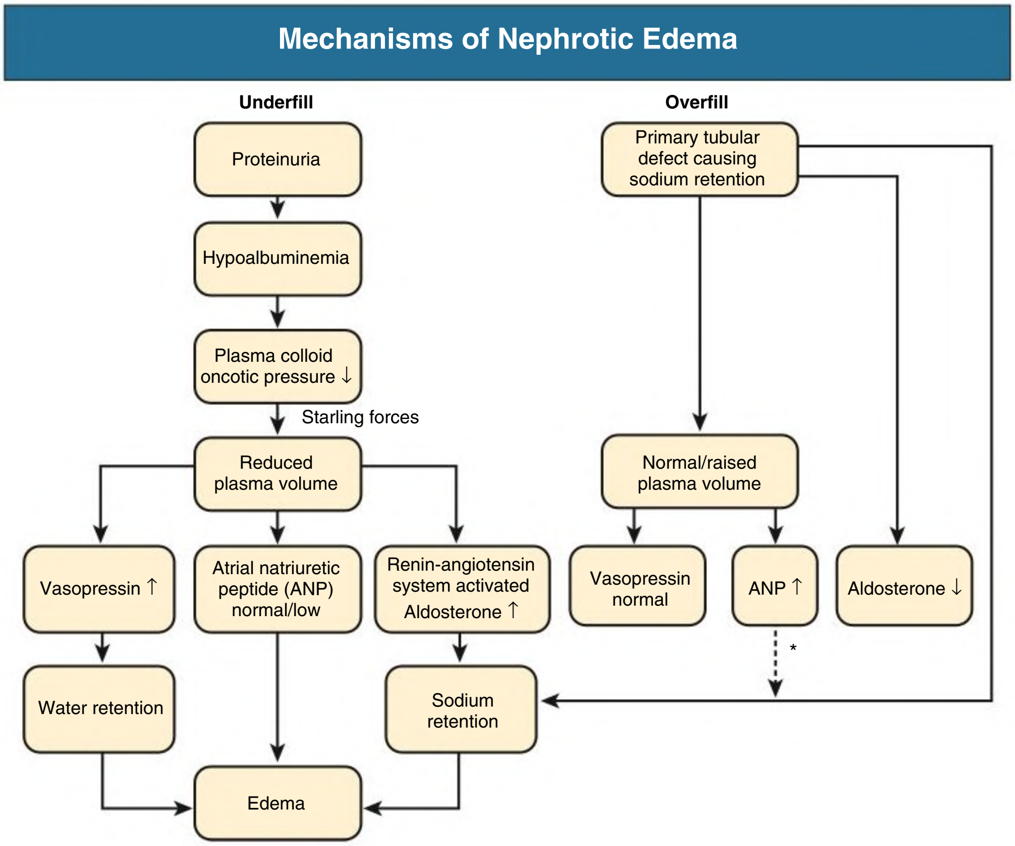

Fig. 16.9 Mechanisms of Nephrotic Edema. *The kidney is relatively resistant to ANP in this setting, so ANP has little effect in countering sodium retention.

(Muehrcke lines) are a characteristic clinical sign of hypoalbuminemia (see Fig. 16.4). The increase in protein synthesis in response to proteinuria is not discriminating; as a result, proteins not being lost in the urine may actually increase in concentration in plasma. This is chiefly determined by molecular weight; large molecules are not lost in urine and will increase in plasma, whereas smaller proteins will enter urine and will decrease in plasma. These variations in plasma proteins are clinically important in two areas: hypercoagulability and hyperlipidemia (see later discussion).

Edema

At least two major mechanisms are involved in the formation of nephrotic edema: underfill and overfill[10] (Fig. 16.9; see Chapter 8). In the first, the edema appears to result from the low serum albumin, producing a decrease in plasma oncotic pressure, which allows increased transudation of fluid from capillary beds into the extracellular space. The consequent decrease in circulating blood volume *(underfill)* results in a secondary stimulation of the renin-angiotensin system (RAS), resulting in aldosterone-induced sodium retention in the distal tubule. This attempt to compensate for hypovolemia merely aggravates edema because the low oncotic pressure alters the balance of forces across the capillary wall in favor of hydrostatic pressure, forcing more fluid into the interstitial space rather than retaining it within the vascular compartment.

However, a much more common mechanism for edema, occurring in most nephrotic patients, is a primary defect in the ability of the distal nephron to excrete sodium, possibly related to activation of the epithelial sodium channel (ENaC) by proteolytic enzymes that enter the tubular lumen in heavy proteinuria.[11] As a result, there is an increased blood volume; suppression of renin, angiotensin, and vasopressin; and a tendency to hypertension rather than to hypotension. The kidney is also relatively resistant to the actions of atrial natriuretic peptide. An elevated blood volume results *(overfill),* which, in association with the low plasma oncotic pressure, provokes transudation of fluid into the extracellular space and edema. In addition to activation of the ENaC (see earlier discussion), it has been hypothesized that inflammatory leukocytes in the interstitium, which are found in many glomerular diseases, may impair sodium excretion by producing angiotensin II and oxidants (oxidants inactivate local nitric oxide, which is natriuretic).[12]

Metabolic Consequences of Nephrotic Syndrome

Negative Nitrogen Balance

The heavy proteinuria leads to marked negative nitrogen balance, usually measured in clinical practice by serum albumin. Nephrotic syndrome is a wasting illness, but the degree of muscle loss is masked by edema and not fully apparent until the patient is rendered edema free. Loss of 10% to 20% of lean body mass can occur. Albumin turnover is increased in response to the tubular catabolism of filtered protein rather than merely to urinary protein loss. Increasing protein intake does not improve albumin metabolism because the hemodynamic response to an increased intake is a rise in glomerular pressure, increasing urine protein losses. A low-protein diet will reduce proteinuria, but it also reduces the albumin synthesis rate and in the longer term may increase the risk for a worsening negative nitrogen balance.

Hypercoagulability

Multiple proteins of the coagulation cascade have altered levels in nephrotic syndrome; in addition, platelet aggregation is enhanced.[13] The net effect is a hypercoagulable state that is enhanced further by immobility, coincidental infection, and hemoconcentration if the

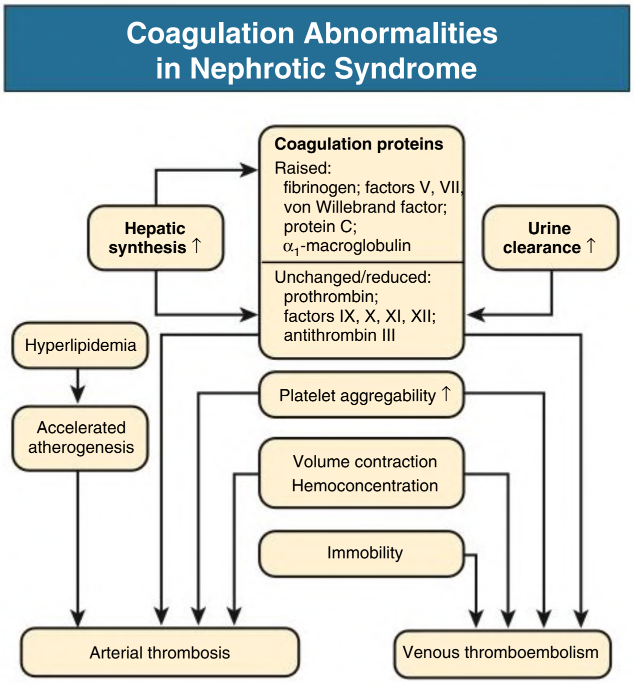

Fig. 16.10 Coagulation abnormalities in nephrotic syndrome.

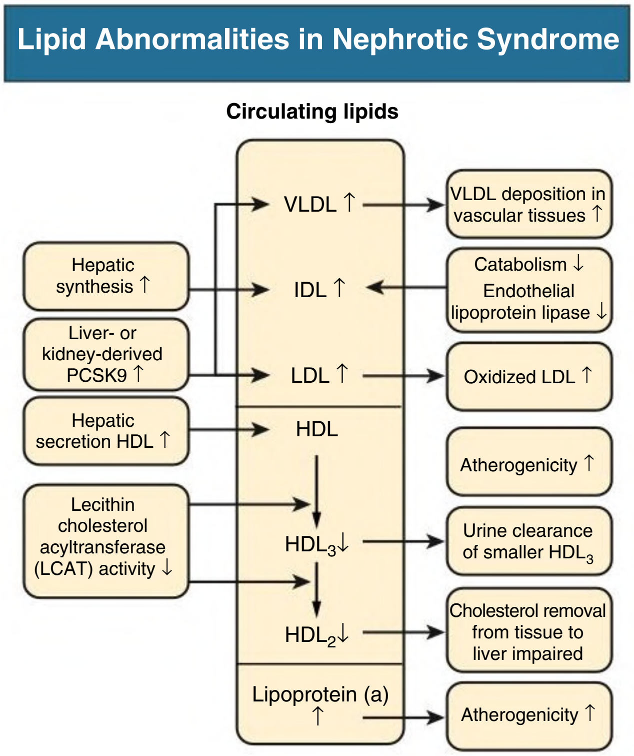

Fig. 16.11 Lipid Abnormalities in Nephrotic Syndrome. Changes in high-density lipoprotein (HDL) are more controversial than those in very-low-density lipoprotein (VLDL). *IDL,* Intermediate-density lipoprotein; *LDL,* low-density lipoprotein; *PCSK9,* proprotein convertase subtilisin/kexin type 9.

patient has a contracted plasma volume (Fig. 16.10). Not only is venous thromboembolism common at any site, but spontaneous arterial thrombosis may rarely occur. Arterial thrombosis may occur in adults, promoting coronary and cerebrovascular events in particular, but it also occurs in nephrotic children, in whom spontaneous thrombosis of major limb arteries can rarely occur. Up to 10% of nephrotic adults and 2% of children will have a clinical episode of thromboembolism. For unexplained reasons, the risk appears particularly high in those with membranous nephropathy.[14] Individual levels of coagulation proteins are not helpful in assessing the risk for thromboembolism, and serum albumin is mostly used as a surrogate marker. Thromboembolic events increase greatly if the serum albumin concentration decreases to less than 2 g/dL.

The hypoproteinemia and dysproteinemia produce a marked increase in erythrocyte sedimentation rate (ESR), which is not a useful marker of an acute phase response in nephrotic patients.

Renal vein thrombosis is an important complication of nephrotic syndrome (see Chapter 43) and presents clinically in up to 8% of nephrotic patients; when sought systematically by ultrasound or contrast venography, the frequency increases to 10% to 50%. Symptoms when the thrombosis is acute may include flank pain and hematuria; rarely, AKI can occur if the thrombosis is bilateral. However, the thrombosis often develops insidiously with minimal symptoms or signs because of the development of collateral blood supply. Pulmonary embolism is an important complication.

Hyperlipidemia and Lipiduria

Hyperlipidemia is so frequently associated with heavy proteinuria that it is regarded as an integral feature of nephrotic syndrome.[15] Clinical stigmata of hyperlipidemia, such as xanthelasmas, may have a rapid onset (see Fig. 16.5). Serum cholesterol concentration can be above 500 mg/dL (13 mmol/L), although serum triglyceride levels are highly variable. The lipid profile in nephrotic syndrome is known to be highly atherogenic in other populations (Fig. 16.11). Many patients who are nephrotic for more than 5 to 10 years will develop additional cardiovascular risk factors, including hypertension and uremia, so it is difficult to separate these influences. However, it is now generally accepted that nephrotic patients have about a fivefold increased risk for coronary death, except for those with MCD, presumably because the nephrotic state is transient before remission with corticosteroid treatment and does not subject the patient with MCD to prolonged hyperlipidemia.

Experimental evidence shows that hyperlipidemia contributes to progressive kidney disease by various mechanisms, with protection afforded by lipid-lowering agents. However, clinical evidence to support a role of statins in slowing CKD progression is inconclusive,[16] so lipid-lowering drugs are indicated in nephrotic syndrome primarily to prevent cardiovascular disease.

Several mechanisms account for the lipid abnormalities in nephrotic syndrome. These include increased hepatic synthesis of low-density lipoprotein (LDL), very-low-density lipoprotein (VLDL), and lipoprotein(a) secondary to the hypoalbuminemia; defective peripheral lipoprotein lipase activity resulting in increased VLDL; and urinary losses of high-density lipoprotein (HDL; see Fig. 16.11). More recently, kidney and hepatic overexpression of proprotein convertase subtilisin/kexin type 9 (PCSK9) has been identified in nephrotic syndrome, which is the basis for a possible new therapeutic intervention using PCSK9 antibodies.[17]

Lipiduria, the fifth component of the nephrotic syndrome, is manifested by the presence of refractile accumulations of lipid in cellular debris and casts (oval fat bodies and fatty casts; Fig. 16.12). However, the lipiduria appears to be a result of the proteinuria and not the plasma lipid abnormalities.

Other Metabolic Effects of Nephrotic Syndrome

Vitamin D–binding protein is lost in the urine, resulting in low plasma 25-hydroxyvitamin D levels, but plasma free vitamin D is usually normal, and overt osteomalacia or uncontrolled hyperparathyroidism is

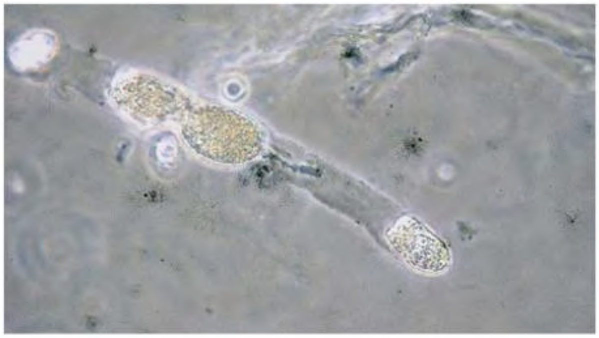

Fig. 16.12 Fat in the Urine. This hyaline cast contains oval fat bodies, which are tubular epithelial cells full of fat. Oval fat bodies often appear brown.

very unusual in nephrotic syndrome in the absence of kidney impairment. Thyroid-binding globulin is lost in the urine and total circulating thyroxine reduced, but again free thyroxine and thyroid-stimulating hormone are normal, and there are no clinical alterations in thyroid status. Occasional patients have been described with copper, iron, or zinc deficiency caused by the loss of binding proteins in the urine.

Drug binding may be altered by the decrease in serum albumin, but most drugs do not require dose modifications. Reduced protein binding also may reduce the dose of vitamin-K antagonists (e.g., warfarin [Coumadin]) required to achieve adequate anticoagulation or the dose of furosemide required to achieve adequate fluid loss (see later discussion).

Infection

Nephrotic patients are prone to bacterial infection. Before corticosteroids were shown to be effective in childhood nephrotic syndrome, sepsis was the most common cause of death, and it remains a major problem in the developing world. Primary peritonitis, especially that caused by pneumococci, is particularly characteristic of nephrotic children. It is less common with increasing age; by 20 years of age, most adults have antibodies against pneumococcal capsular antigens. Peritonitis caused by both β-hemolytic streptococci and gram-negative organisms occurs, but staphylococcal peritonitis is not reported. Cellulitis, especially in areas of severe edema, is also common, most frequently caused by β-hemolytic streptococci.

The increased risk for infection has several explanations. Large fluid collections are sites for bacteria to grow easily; nephrotic skin is fragile, creating sites of entry; and edema may dilute local humoral immune factors. Loss of immunoglobulin G (IgG) and complement factor B (of the alternative pathway) in the urine impairs host ability to eliminate encapsulated organisms such as pneumococci. Zinc and transferrin are lost in the urine, and both are required for normal lymphocyte function. Neutrophil phagocytic function is impaired in patients with nephrotic syndrome, and several forms of in vitro T cell dysfunction are described, although their clinical significance is unclear.

Acute and Chronic Changes in Kidney Function

Acute Kidney Injury

Patients with nephrotic syndrome are at risk for AKI[18] through a variety of mechanisms (Box 16.1). These include volume depletion or sepsis, resulting in prerenal AKI or acute tubular necrosis[19]; transformation of the underlying disease, such as crescentic nephritis in a patient with membranous nephropathy; bilateral renal vein thrombosis; increased disposition to prerenal AKI from NSAIDs and ACE inhibitors or ARBs; and increased risk for allergic interstitial nephritis secondary to drugs, including diuretics. AKI may rarely result from intrarenal edema with compression of tubules and, as with nephrotic patients with prerenal azotemia, may respond with diuresis to albumin infusions combined with a loop diuretic.

BOX 16.1 Causes of Acute Kidney Injury in Nephrotic Syndrome

- Prerenal failure caused by volume depletion
- Acute tubular necrosis caused by volume depletion and/or sepsis
- Intrarenal edema
- Renal vein thrombosis
- Transformation of underlying glomerular disease (e.g., crescentic nephritis superimposed on membranous nephropathy)
- Adverse effects of drug therapy
- Acute allergic interstitial nephritis secondary to various drugs, including diuretics
- Hemodynamic response to NSAIDs and ACE inhibitors or ARBs

ACE, Angiotensin-converting enzyme; *ARBs,* angiotensin receptor blockers; *NSAIDs,* nonsteroidal antiinflammatory drugs.

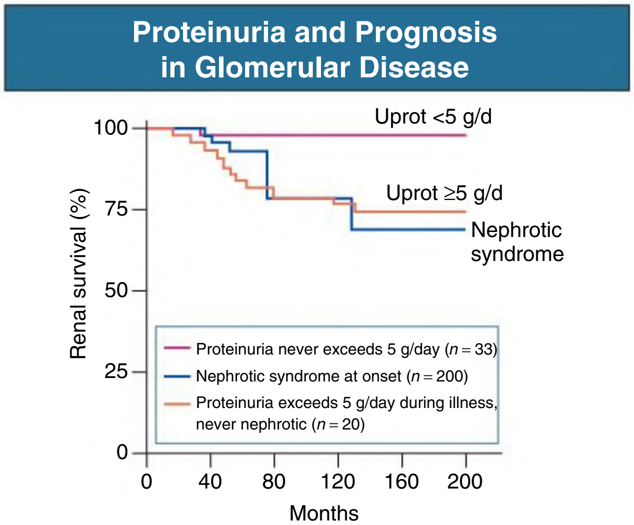

Fig. 16.13 Proteinuria and Prognosis in Glomerular Disease. The influence of heavy proteinuria *(Uprot)* on long-term renal function in 253 patients with primary glomerular disease at Manchester Royal Infirmary, United Kingdom. Heavy proteinuria at any time during long-term follow-up substantially worsens the prognosis even without frank nephrotic syndrome. (Courtesy Dr. C.D. Short.)

Chronic Kidney Disease

With the exception of MCD, most causes of nephrotic syndrome are associated with progressive kidney failure. A major risk factor for progression is the degree of proteinuria (see Chapters 81 and 82). Progression is uncommon if there is sustained proteinuria of less than 2 g/day. The risk increases in proportion to the severity of the proteinuria, with marked risk for progression when protein excretion is more than 5 g/day (Fig. 16.13). Proteinuria identifies patients with severe glomerular injury; however, experimental and clinical evidence also suggests that proteinuria itself may be toxic, especially to the tubulointerstitium.[20] In experimental models, measures that reduce proteinuria (e.g., ACE inhibitors) also prevent tubulointerstitial disease and progressive kidney failure.

NEPHRITIC SYNDROME

In *nephrotic* syndrome, the glomerular injury is manifested primarily as an increase in permeability of the capillary wall to protein. By contrast,

TABLE 16.4 Differentiation Between Nephrotic Syndrome and Nephritic Syndrome

Typical Features	Nephrotic	Nephritic
Onset	Insidious	Abrupt
Edema	++++	++
Blood pressure	Normal	Raised
Jugular venous pressure	Normal/low	Raised
Proteinuria	++++	++
Hematuria	May/may not occur	+++
Red blood cell casts	Absent	Present
Serum albumin	Low	Normal/slightly reduced

TABLE 16.5 Common Glomerular Diseases Presenting as Nephritic Syndrome

Disease	Associations	Serologic Tests
Poststreptococcal glomerulonephritis	Pharyngitis, impetigo	Antistreptolysin titer, streptozyme antibody
Other postinfectious disease		
Endocarditis	Cardiac murmur	Blood cultures, C3↓
Abscess	—	Blood cultures, C3, C4 normal or increased
Shunt	Treated hydrocephalus	Blood cultures, C3↓
IgA nephropathy	Upper respiratory or gastrointestinal infection	Serum IgA↑
Lupus nephritis	Other multisystem features of lupus	Antinuclear antibody, anti–double-stranded DNA antibody, C3↓, C4↓

IgA, Immunoglobulin A.

in *nephritic* syndrome, there is evidence of glomerular inflammation resulting in a reduction in GFR, nonnephrotic proteinuria, edema and hypertension (secondary to sodium retention), and hematuria with RBC casts.

The classic nephritic syndrome presentation is seen with acute poststreptococcal GN in children, but this complication has become rare. These children usually present with rapid onset of oliguria, weight gain, and generalized edema over a few days. The hematuria results in brown rather than red urine, and clots are not seen. The urine contains protein, RBCs, and RBC casts. Because proteinuria is rarely in the nephrotic range, serum albumin concentration is usually normal. Circulating volume increases with hypertension, and pulmonary edema follows without evidence of primary cardiac disease.

The distinction between typical nephrotic syndrome and nephritic syndrome is usually straightforward on clinical and laboratory grounds (Table 16.4). The use of these clinical descriptions for patients with suspected GN at first presentation helps narrow the differential diagnosis. However, the classification systems are imperfect, and patients with certain glomerular disease patterns, such as a membranoproliferative GN pattern on biopsy, may present with nephrotic syndrome, nephritic syndrome, or a combination of both syndromes.

Etiology

Table 16.5 lists the primary glomerular diseases associated with nephritic syndrome and the serologic tests helpful in diagnosis. The classification is even more challenging than for nephrotic syndrome, because some diseases are identified by histology (IgA nephropathy), some by serology and histology (ANCA-associated vasculitis and lupus nephritis), and others by etiology (postinfectious GN).

RAPIDLY PROGRESSIVE GLOMERULONEPHRITIS

In rapidly progressive glomerulonephritis (RPGN), glomerular injury is so acute and severe that kidney function deteriorates markedly over days or weeks. The patient may present as a uremic emergency, with nephritic syndrome that rapidly progresses to kidney failure, or with rapidly deteriorating kidney function when being investigated for extrarenal disease (many of the GN associated with RPGN occur as part of a systemic immune illness).

The histologic counterpart of RPGN is crescentic GN. The proliferative cellular response seen outside the glomerular tuft but within Bowman's space is known as a "crescent" because of its shape on histologic cross section (see Fig. 17.8). Typically, the glomerular tuft also shows segmental necrosis or focal segmental necrotizing GN, especially in the vasculitis syndromes.

The term *rapidly progressive* GN is therefore often used to describe acute deterioration in kidney function in association with crescentic nephritis. Unfortunately, not all patients with a nephritic urine sediment and AKI will fit this syndrome. For example, AKI may also occur in milder glomerular disease if it is complicated by accelerated hypertension, renal vein thrombosis, or acute tubular necrosis. This emphasizes the need to obtain histologic confirmation of the clinical diagnosis.

Etiology

Table 16.6 shows the primary glomerular diseases associated with RPGN and helpful serologic tests. As with nephritic syndrome, different assessment methods are useful for different diseases causing RPGN.

PROGRESSIVE CHRONIC KIDNEY DISEASE

In most types of chronic GN, between 25% and 50% of patients will have slowly progressive kidney impairment. Patients may present late with established hypertension, proteinuria, and kidney impairment. In long-standing GN, the kidneys shrink but remain smooth and symmetric. Kidney biopsy at this stage is more hazardous and less likely to provide diagnostic material. Light microscopy often shows nonspecific features of end-stage kidney disease (ESKD), consisting of focal or global glomerulosclerosis and dense tubulointerstitial fibrosis, and it may not be possible to confirm that a glomerular disease was responsible, let alone define the glomerular pattern more precisely. Immunofluorescence may be more helpful; in particular, mesangial IgA may be present in adequate amounts to diagnose IgA nephropathy. However, when kidney imaging shows small kidneys, only rarely will biopsy be appropriate, and so chronic GN often has been a presumptive diagnosis in patients presenting late with shrunken kidneys, proteinuria, and kidney impairment. This in turn may have led to an overestimate of the frequency of GN as a cause of ESKD in registry data. GN should be diagnosed only if there is confirmatory histologic evidence.

TABLE 16.6 Glomerular Diseases Presenting as Rapidly Progressive Glomerulonephritis

Disease	Associations	Serologic Tests
Goodpasture Syndrome		
	Lung hemorrhage	Anti–glomerular basement membrane antibody (occasionally ANCA present)
Vasculitis		
Granulomatosis with polyangiitis	Upper and lower respiratory tract involvement	Mostly PR3 ANCA
Microscopic polyangiitis	Multisystem involvement	Mostly MPO ANCA
Pauci-immune crescentic glomerulonephritis	Renal involvement only	MPO or PR3 ANCA
Immune Complex Disease		
Systemic lupus erythematosus	Other multisystem features of lupus	Antinuclear antibody, anti–double-stranded DNA antibody, C3↓, C4↓
Poststreptococcal glomerulonephritis	Pharyngitis, impetigo	Antistreptolysin titer, streptozyme antibody C3↓, C4 normal
IgA nephropathy; IgA vasculitis (HSP)	Characteristic rash ± abdominal pain in IgA vasculitis	Serum IgA↑(30%) C3 and C4 normal
Endocarditis		
	Cardiac murmur; other systemic features of bacteremia	Blood cultures ANCA (occasionally) C3↓, C4 normal

Note the overlap between these diseases and those in Table 16.5. A number of glomerular diseases may present with either nephritic syndrome or RPGN.
ANCA, Antineutrophil cytoplasmic antibody; *HSP,* Henoch-Schönlein purpura; *IgA,* immunoglobulin A; *MPO,* myeloperoxidase; *PR3,* proteinase 3; *RPGN,* rapidly progressive glomerulonephritis.

TREATMENT OF GLOMERULAR DISEASE

General Principles

Before any therapeutic decisions, it should always be ascertained that glomerular disease is *primary* and that no specific therapy is indicated. For example, treatment of an underlying infection or tumor may result in remission of GN. In the remaining cases, both general supportive treatment (see Chapter 82) and disease-specific therapy should be considered. Supportive treatment includes treatment of hypertension, proteinuria, edema, and other metabolic consequences of nephrotic syndrome. If successful, these relatively nontoxic therapies can prevent the need for immunosuppressive drugs, which have multiple potential side effects. Supportive therapy is usually not necessary in corticosteroid-sensitive MCD with rapid remission or in patients with IgA nephropathy, Alport syndrome, or thin basement membrane nephropathy, provided the patient exhibits no proteinuria above 0.5 g/day, loss of GFR, or hypertension.

Hypertension

Hypertension is very common in GN; it is virtually universal as chronic GN progresses toward kidney failure and is the key modifiable factor in preserving kidney function. Sodium and water overload is an important part of the pathogenetic process, and high-dose diuretics with moderate dietary sodium restriction are usually an essential part of the treatment. As in other chronic kidney diseases, the blood pressure control reduces cardiovascular risk and also delays progressive kidney function loss. In the Modification of Diet in Renal Disease (MDRD) study, patients with proteinuria (>1 g/day) had a better outcome if their blood pressure was reduced to 125/75 mm Hg rather than to the previous standard of 140/90 mm Hg.[21,22] The recent Kidney Disease: Improving Global Outcomes (KDIGO) CKD guideline recommends a blood pressure target of 130/80 mm Hg in proteinuric patients,[23] but the SPRINT trial and some other studies suggest that a systolic blood pressure target in the 120s may reduce cardiovascular complications in nondiabetic subjects compared with a higher target (140 mm Hg; see Chapters 35 and 82). Clinical trial data strongly supports the use of ACE inhibitors and ARBs as the first-choice therapy.[24-26] Nondihydropyridine calcium channel blockers may also have a beneficial effect on proteinuria as well as on blood pressure. In contrast, dihydropyridine calcium channel blockers may exacerbate proteinuria because of their ability to dilate the afferent arteriole, but these agents are considered relatively safe to use if the patient is receiving either an ACE inhibitor or an ARB. As in primary hypertension, lifestyle modification (salt restriction, weight normalization, regular exercise, and smoking cessation) should be an integral part of the therapy.[23] If target blood pressure cannot be achieved with these measures, antihypertensive therapy should be stepped up according to current guidelines (see Chapter 37).

Treatment of Proteinuria

Besides hypertension, proteinuria represents the second key modifiable factor to preserve GFR in patients with glomerular disease (see Chapters 81 and 82). Most studies suggest that progressive loss of kidney function can be minimized if proteinuria can be reduced to levels below 0.5 g/d. This may be because many of the measures to reduce protein excretion (e.g., ACE inhibitors, ARBs) also reduce glomerular hypertension, which contributes to progressive kidney failure. However, there is also increasing evidence that proteinuria or factors present in proteinuric urine may be toxic to the tubulointerstitium.[23] In nephrotic patients, a reduction of proteinuria to a nonnephrotic range can induce serum proteins to rise, with alleviation of many of the metabolic complications of nephrotic syndrome.

Most of the agents used to reduce urinary protein excretion do so by blocking efferent arteriolar constriction (ACE inhibitors or ARBs) or reducing preglomerular pressure (most other classes of antihypertensive drugs). As mentioned, dihydropyridine calcium antagonists are the exception because they preferentially dilate the afferent arteriole and thereby can increase intraglomerular pressure and thus exacerbate

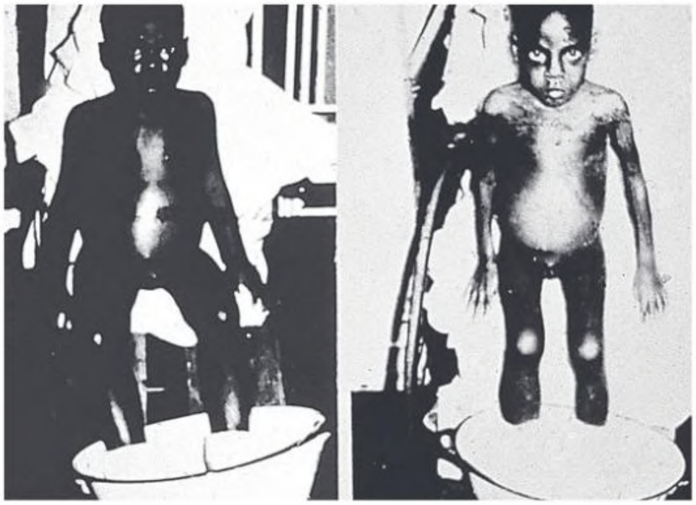

Fig. 16.14 **Treatment of Nephrotic Edema before the Availability of Diuretics.** Edema in nephrotic syndrome was very difficult to treat. In 1953 this child with anasarca stands in a bowl while edema fluid drips out through small tubes placed through needles in the skin of the feet. Nevertheless, this was effective treatment. The two pictures of the same child were taken 4 days apart, during which time the child lost 4.5 kg (10 lb), or 18% of body weight. (Courtesy Dr. Robert Vernier.)

proteinuria. Some of the agents, such as ACE inhibitors and ARBs, also may directly reduce the increased glomerular capillary wall permeability. A consequence of this type of therapy is a reduction in GFR; in general, however, the decrease in GFR is of a lower magnitude than the decrease in protein excretion. ACE inhibitors and ARB reduce proteinuria by an average of 40% to 50%, particularly if the patient is on dietary salt restriction. There is little clinical evidence to suggest that ACE inhibitors differ from ARBs in this respect. The combination of ACE inhibitors and ARBs does not appear to provide additional benefit and increases the risk for AKI and hyperkalemia, in particular when used in older people with vascular disease and diabetes.[27,28] Similar concerns relate to the combination of an ACE inhibitor or ARB with a direct renin inhibitor.[29]

In addition, whereas other classes of antihypertensive agents will reduce proteinuria coincident with a decrease in systemic blood pressure, particularly the nondihydropyridine calcium channel blockers such as diltiazem, both ACE inhibitors and ARBs usually reduce proteinuria independent of blood pressure. If doses are increased slowly to minimize symptomatic hypotension, treatment with ACE inhibitors and ARBs is usually possible in the normotensive proteinuric patient. Common side effects include hyperkalemia in patients with advanced CKD, which may necessitate a loop diuretic but rarely should lead to cessation of ACE inhibitors and ARBs, and cough with ACE inhibitors, in which case ARBs should be used instead. Because both agents lower GFR, a 10% to 30% increase in serum creatinine concentration may be observed. Unless serum creatinine continues to increase, this moderate increase reflects the therapeutic effect of ACE inhibitors and ARBs and should not prompt their withdrawal. Finally, if proteinuria persists despite maximum allowed or tolerated doses of ACE inhibitors or ARBs, a low dose of an aldosterone antagonist may overcome so-called aldosterone breakthrough and further reduce proteinuria (see Chapter 82). Hyperkalemia is an important safety aspect, similar to the ACE inhibitor and ARB combination mentioned earlier.

The NSAIDs lessen proteinuria by reducing intrarenal prostaglandin production and dipyridamole through adenosine-mediated afferent arteriolar vasoconstriction. However, they are generally contraindicated as their use can cause profound decreases in GFR, salt retention, and diuretic resistance.

A low-protein diet will lessen proteinuria but must be advised with great care because of the risk for malnutrition.[23] Adequate compensation must be made for urine protein losses,[30] and the patient must be carefully monitored for evidence of malnutrition (see Chapter 90). Whether a low-protein diet is still antiproteinuric in patients treated with a full dose of ACE inhibitor or ARB is not established.

Treatment of Hyperlipidemia

Treatment of hyperlipidemia (or hypercholesterolemia) in patients with glomerular disease should usually follow the guidelines that apply to the general population to prevent cardiovascular disease. A statin or statin/ezetimibe combination is recommended in adults older than 50 years with CKD stage 3 to 5. Statins alone are also recommended in adults over 50 years with earlier-stage CKD. In younger adults, statins should be considered if the patient has significant comorbidity (coronary disease, diabetes mellitus, stroke; see Chapter 85). Statins may protect from a decrease in GFR, although this is not firmly established. Dietary restriction alone has only modest effects on hyperlipidemia in glomerular disease, particularly nephrotic syndrome. Side effects of some medications, such as rhabdomyolysis provoked by fibrates, occur more frequently in patients with kidney failure.

Avoidance of Nephrotoxic Substances

Apart from NSAIDs, which may induce AKI, particularly in patients with preexisting kidney impairment and dehydration, other nephrotoxic substances, such as radiocontrast agents, some cytotoxic drugs, and antibiotics (e.g., aminoglycosides), also should be used with caution in patients with glomerular disease and kidney impairment or nephrotic syndrome. There is also increasing concern that proton pump inhibitors may be associated with increased risk for both acute interstitial nephritis and CKD, and thus their use should be restricted to documented indications when histamine-2 blockers are not effective.

Special Therapeutic Issues in Patients With Nephrotic Syndrome

Treatment of Nephrotic Edema

In contrast to the lack of therapies in the past (Fig. 16.14), the mainstay of current treatment of nephrotic edema is diuretic therapy accompanied by moderate dietary sodium restriction (60–80 mmol/24 h).

Nephrotic patients are diuretic resistant even if GFR is normal. Loop diuretics must reach the kidney tubule to be effective, and transport from the peritubular capillary requires protein binding, which is reduced in hypoalbuminemia. When the drug reaches the kidney tubule, it will become 70% bound to protein present in the urine and therefore be less effective. Oral diuretics with twice-daily administration are usually preferred, given the longer therapeutic effect compared with intravenous diuretics. However, in severe nephrosis, GI absorption of the diuretic may be uncertain because of intestinal wall edema, and intravenous diuretic by bolus injection or infusion may be necessary to provoke an effective diuresis. Alternatively, combining a loop diuretic with a thiazide diuretic or with metolazone may overcome diuretic resistance (see Chapter 8). A third alternative is adding amiloride to loop diuretics because nephrotic syndrome is associated with activation of the ENaC channel. Significant hypovolemia is not often a clinical problem, provided fluid removal is controlled and gradual. Daily weight should decrease by no more than 1 to 2 kg/day. Nephrotic children are much more prone to hypovolemic shock than adults. A stepwise approach to diuretic use is required, aiming at fluid removal in adults of no more than 2 kg daily, moving on to the next drug level if this is not achieved (Fig. 16.15).

Correction of Hypoproteinemia

Adequate dietary protein should be ensured in nephrotic patients (0.8–1 g/kg/day) with a high carbohydrate intake to maximize use of that protein. In patients with heavy proteinuria, the amount of urinary protein loss should be added to dietary protein intake.

In the very rare setting of proteinuria so severe that the patient is dying of the complications of nephrotic syndrome, the physician may need to resort to medical nephrectomy to prevent continued protein losses: the deliberate use of NSAIDs combined with ACE inhibitors and diuretic to lessen proteinuria by provoking AKI. If medical nephrectomy alone does not adequately reduce proteinuria, bilateral renal artery embolization can be considered. It is a painful procedure and is not always as successful (perhaps because of collateral arterial supply to the kidneys, which is not blocked by embolization). A final alternative is bilateral nephrectomy, which can be performed by laparoscopic surgery with reduced morbidity compared with classical surgical removal. Bilateral nephrectomy is commonly used in the management of infants with congenital nephrotic syndrome.

Treatment of Hypercoagulability

The risk for thrombotic events increases as serum albumin values decrease to less than 2.5 g/dL. Immobility as a consequence of edema or intercurrent illness further aggravates the risk. Prophylactic low-dose anticoagulation (e.g., heparin 5000 U subcutaneously twice daily) is indicated at times of high risk, such as relative immobilization in the hospital and albumin levels between 2 and 2.5 g/dL. Full-dose anticoagulation with low-molecular-weight heparin or warfarin should be considered if serum albumin decreases to less than 2 g/dL,[13,31] and this is mandatory if a thrombosis or pulmonary embolism is documented. Heparin is used for initial anticoagulation, but an increased dose may be needed because part of the action of heparin depends on circulating antithrombin III, which is often reduced in nephrotic patients. Warfarin (target international normalized ratio [INR] 2 to 3) is the long-term treatment of choice but should be adjusted with special care because of altered protein binding, which may require dose reductions. Newer oral anticoagulants have not been systematically tested so far in this situation.

Management of Infection

A high degree of clinical suspicion for infection is vital in nephrotic patients. Especially in nephrotic children, ascitic fluid should be examined microscopically and cultured if there is any suspicion of systemic infection. Bacteremia is common even if clinical signs are localized. ESR is unhelpful, but an elevated C-reactive protein level may be informative. Parenteral antibiotics should be started once culture specimens are taken, and the regimen should include coverage for pneumococci. If repeated infections occur, serum immunoglobulins should be measured. If serum IgG is less than 600 mg/dL, an uncontrolled study suggested that infection risk is reduced by monthly administration of intravenous immune globulin (10–15 g) to keep the IgG levels above 600 mg/dL.[32]

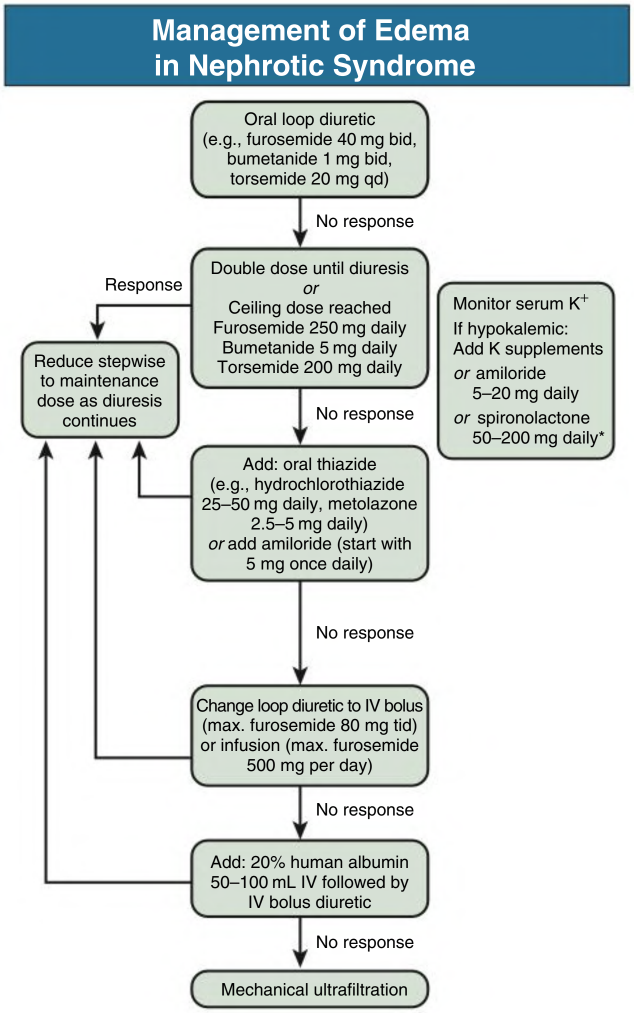

Fig. 16.15 **Management of Edema in the Patient With Nephrotic Syndrome.** Edema is often diuretic resistant, but the response is not predictable. Therefore, stepwise escalation of therapy is appropriate until diuresis occurs. Even when there is anasarca, diuresis should not proceed faster than 2 kg/day in adults to minimize the risk for clinically significant hypovolemia. Mechanical ultrafiltration is rarely required for nephrotic edema unless there is associated renal impairment. *Spironolactone is less effective in nephrotic syndrome than in cirrhosis and is often poorly tolerated because of gastrointestinal side effects. Spironolactone should be used with great caution if the glomerular filtration rate is very low. *bid,* Twice daily; *IV,* intravenous.

Disease-Specific Therapies

Specific treatments for glomerular diseases are discussed in the subsequent chapters; the general principles are discussed here. Most glomerular diseases have an immune pathogenesis, and treatment has generally consisted of immunosuppressive therapy. In the patient with glomerular disease resulting from ineffectual elimination of a foreign antigen, treatment aims at eliminating this antigen, such as antibiotics in endocarditis-associated GN or antiviral therapy for cryoglobulinemia resulting from hepatitis C viral infection.

In general, the more severe and acute the presentation of GN, the more successful is immunosuppressive treatment. Immunosuppression in chronic GN has had minimal success. When kidney function is declining rapidly, as in RPGN, the toxicity of intensive regimens becomes acceptable for a short period, although it would be unacceptable if prolonged. Furthermore, the nonspecific nature of most immunosuppressive treatments results in widespread interruption of immune and inflammatory events at multiple levels. Despite great increases in the understanding of immune mechanisms in glomerular disease since the 1970s, most immunosuppressive therapies are not yet much more specific or precise. The mainstays of treatment include corticosteroids, cyclosporine, tacrolimus, cyclophosphamide, mycophenolate mofetil, and/or rituximab. Novel agents, including some that modify inflammation (e.g., complement antagonists) or B cell immunity (e.g., inhibitors of B cell activation factor), are now being studied in clinical trials in glomerular disease but as yet have no proven indications.

The use of immunosuppressive therapies to treat patients with GN has drawbacks. In many diseases, good prospective controlled trials are often lacking. If sufficient glomerular damage is present, proteinuria and progressive deterioration of kidney function may occur by nonimmune pathways that may not be responsive to immunosuppressive therapies. This is particularly relevant in patients in whom the GN has already resulted in advanced CKD. Unfortunately, good noninvasive markers to assess disease activity are missing in most clinical circumstances. Given the frequent uncertainty of the response to immunosuppressive therapy, it becomes mandatory to weigh the potential benefits against the risks of therapy.

Immunosuppression may be associated with reactivation of tuberculosis and hepatitis B infection and also can lead to a hyperinfection syndrome in patients with *Strongyloides* infection. Therefore, high-risk patients should be tested for these diseases before embarking on therapy.

Alkylating agents such as cyclophosphamide have considerable toxicity. In the short term, leukopenia is common, as is alopecia, although hair will regrow within a few months with discontinuation of therapy. These agents can cause infertility (observed in adults with cumulative doses of cyclophosphamide >200 mg/kg). There is also an increased incidence of leukemias (observed with total doses of cyclophosphamide >80 g). Cyclophosphamide is also a bladder irritant, and treatment can result in hemorrhagic cystitis and bladder carcinoma, particularly after therapy lasting more than 6 months.[33] Irritation of the bladder is caused by a metabolite, acrolein. The effect can be minimized in patients receiving intravenous cyclophosphamide by enforcing a good diuresis and administering mesna. The dose of mesna (milligrams) should equal the dose of cyclophosphamide (milligrams); 20% is given intravenously with the intravenous cyclophosphamide, and the remaining 80% should be given in two equal oral doses at 2 and 6 hours. Cyclophosphamide also requires dose reduction in the setting of impaired kidney function. Given all these concerns, oral treatment with cyclophosphamide should ideally be limited to 12 weeks.

The modes of action and potential adverse effects of corticosteroids, azathioprine, and other immunosuppressive agents occasionally used in glomerular disease are discussed further in Chapter 106.

SELF-ASSESSMENT QUESTIONS

1. Which of the following statements regarding proteinuria is *correct?*
   A. Tubular proteinuria is characterized by equal amounts of α_1-microglobulin and immunoglobulins.
   B. Proteinuria exceeding 3.5 g/day (i.e., "nephrotic" proteinuria) inevitably results in hypoalbuminemia.
   C. In orthostatic proteinuria, urinary protein is typically increased with the patient lying down.
   D. Overflow proteinuria is typical of urinary light chain excretion.
   E. Functional proteinuria typically occurs after heavy meals with high protein intake.
2. Which of the following statements regarding general treatment of glomerular disease is *incorrect*?
   A. The KDIGO CKD guideline recommends a blood pressure target below 130/80 mm Hg in proteinuric patients.
   B. ACE inhibitors and ARBs may directly reduce the increased glomerular capillary wall permeability.
   C. Common side effects of ACE inhibitors and ARBs in patients with advanced CKD include hyperkalemia, which may necessitate a loop diuretic.
   D. A statin or a statin-ezetimibe combination is recommended in all adults younger than 50 years with CKD.
   E. A low-protein diet will lessen proteinuria but must be advised with great care because of the risk of malnutrition.
3. Which statement regarding the treatment of nephrotic syndrome is *correct?*
   A. Nephrotic patients are diuretic resistant only if GFR is greatly impaired.
   B. Patients with heavy proteinuria should be advised to consume a high-protein diet of about 2 to 3 g/kg/day.
   C. Full-dose anticoagulation with low-molecular-weight heparin or warfarin should be considered if serum albumin decreases to less than 2 g/dL.
   D. Ascitic fluid should be examined regularly by microscopy and culture independently of a suspicion of systemic infection.
   E. Nephrotic children are no more prone to hypovolemic shock than are adults.

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